MambaGuard-MOO: Multi-Objective Safe Drug Recommendation with Contrastive Learning

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***Abstract*—Drug recommendation systems in healthcare face the fundamental challenge of simultaneously optimizing multiple critical objectives including therapeutic efficacy, safety constraints, dosage appropriateness, and toxicity management, which cannot be adequately addressed by simple language model-based methods. To tackle this multi-objective optimization challenge, we propose MambaGuard-MOO, a novel framework that enhances the MambaDrug architecture with advanced contrastive learning mechanisms for safe and effective medication recommendation. Unlike conventional approaches that treat drug recommendation as a single-objective problem, our method explicitly models the complex trade-offs between treatment effectiveness and patient safety through a multi-objective optimization framework. MambaGuard-MOO integrates Drug-Drug Interaction (DDI) constraints, dosage considerations, and toxicity profiles while leveraging contrastive learning to better distinguish between safe and unsafe drug combinations. Our approach dynamically balances multiple objectives including therapeutic efficacy, DDI minimization, dosage optimization, and adverse effect reduction through an adaptive weighting mechanism. Extensive experiments on the MIMIC-III dataset demonstrate the effectiveness of our approach in achieving superior performance across multiple safety and efficacy objectives, establishing a new paradigm for safe and effective drug recommendation systems.**

***Index Terms*—drug recommendation, multi-objective optimization, contrastive learning, Mamba architecture, healthcare AI**

I. Introduction

The development of precise drug recommendation systems represents a critical frontier in modern healthcare, with the potential to dramatically improve patient outcomes and reduce medical errors globally. Healthcare professionals face unprecedented challenges navigating complex pharmaceutical landscapes, where many deaths are caused by medical errors in hospitals each year, many stemming from inappropriate medication prescriptions. The overload of medical information on drugs, medical tests, and treatment suggestions has brought many difficulties to medical professionals in making patient-oriented decisions [1]. A sophisticated drug recommendation system that simultaneously optimizes therapeutic efficacy while ensuring patient safety could transform clinical decision-making and ultimately save lives by providing intelligent, data-driven guidance for medication selection.

Despite significant advances in computational approaches to drug recommendation, existing methodologies suffer from fundamental limitations that constrain their effectiveness. Current state-of-the-art systems such as Cycletrans [2] and medGPT [3] primarily focus on isolated aspects of the recommendation problem, typically emphasizing either Drug-Drug Interaction (DDI) minimization or symptom-to-medication mapping through single-objective optimization frameworks. Current recommendations are manually prepared by physicians, but this is difficult for patients with multiple comorbidities [4]. The predominant single-objective focus leads to suboptimal trade-offs, where improvements in one aspect often come at the expense of others, resulting in recommendations that may be mathematically optimal for specific metrics but clinically suboptimal for comprehensive patient care.

To address these limitations, we propose MambaGuard-MOO, a novel multi-objective optimization framework that enhances the MambaDrug architecture through advanced contrastive learning mechanisms and comprehensive safety constraints. Our approach represents a paradigm shift from single-objective to multi-objective optimization, explicitly modeling the complex trade-offs inherent in clinical decision-making through simultaneous optimization of therapeutic efficacy, DDI minimization, dosage optimization, and toxicity reduction. The incorporation of contrastive learning enables our system to better distinguish between safe and unsafe drug combinations by learning rich representations that capture critical differences in drug interaction patterns [5]. Our multi-objective optimization framework employs adaptive weighting mechanisms that dynamically balance competing objectives based on patient-specific factors, ensuring recommendations are both mathematically optimal and clinically implementable.

II. Related Work

A. Drug Recommendation Systems

Drug recommendation systems have evolved significantly from traditional rule-based approaches to sophisticated deep learning architectures [4]. Early drug recommendation methods relied primarily on collaborative filtering and content-based approaches [1]. Recent advances in deep learning have led to the development of neural network-based drug recommendation systems that can model complex drug-patient interactions and predict potential adverse effects [5]. Deep neural network-assisted drug recommendation systems have

demonstrated significant improvements in identifying drug-target interactions through advanced architectures such as bi-directional LSTM and graph convolutional networks [6]. Contemporary approaches like SafeDrug [7] leverage dual molecular graph encoders to capture both global pharmacological properties and local sub-structural patterns of drugs, while explicitly modeling drug-drug interactions through controllable loss functions. Knowledge graph-driven approaches have also emerged as powerful frameworks for medicine recommendation, utilizing graph neural networks to learn embeddings from comprehensive medical knowledge graphs constructed from longitudinal electronic health records [8]. Multi-modal fusion approaches incorporate contrastive learning mechanisms to enhance drug-target interaction prediction by learning rich representations from diverse biomedical data sources [9].

B. State Space Models and Mamba Architecture

State Space Models (SSMs) have emerged as a promising alternative to transformer architectures for sequence modeling, offering linear computational complexity while maintaining the ability to capture long-range dependencies [10]. The Mamba architecture, built upon selective structured state space models, addresses key limitations of previous SSMs by incorporating selective mechanisms that allow models to selectively propagate or forget information based on the current input, enabling content-based reasoning [10]. This selective capability makes Mamba particularly suitable for healthcare applications where different parts of a patient's medical history may have varying relevance for current treatment decisions. Recent applications of Mamba to recommendation systems, such as Mamba4Rec [11], have demonstrated the architecture's effectiveness in sequential recommendation tasks, achieving superior performance compared to both RNN and attention-based baselines while maintaining computational efficiency [11]. The linear scaling properties of Mamba architectures make them particularly attractive for processing long medical sequences, thus opening new possibilities for comprehensive patient history modeling in drug recommendation systems. Various extensions of state space models, including structured architectures and gating mechanisms, have been developed to enhance their capabilities across different domains [12].

III. Problem Formulation

We formulate the drug recommendation problem as a multi-objective sequential prediction task that leverages comprehensive electronic health records (EHR) while ensuring medication safety through drug-drug interaction (DDI) constraints. Our problem builds upon the MambaDrug framework and extends it to address the complex trade-offs inherent in clinical decision-making.

A. Input Data Representation

Given a patient's longitudinal medical history, we represent each visit as a triple $x^{(t)} = \{d^{(t)}, p^{(t)}, m^{(t)}\}$, where $d^{(t)} \in \{0,1\}^{|D|}$ represents a multi-hot diagnosis vector, $p^{(t)} \in \{0,1\}^{|P|}$ represents a multi-hot procedure vector, and $m^{(t)} \in \{0,1\}^{|M|}$ represents the prescribed medications at visit t. Here, D, P, and M denote the sets of all possible diagnoses, procedures, and medications, respectively. For a patient with T visits, the complete medical history is represented as $X = \{x^{(1)}, x^{(2)}, \ldots, x^{(T)}\}$.

B. Safety Constraints

To ensure medication safety, we incorporate two critical constraint matrices. The drug-drug interaction matrix $\mathbf{A}_{DDI} \in \{0,1\}^{|M|\times|M|}$ captures known adverse interactions between medication pairs, where $\mathbf{A}_{DDI}[i,j] = 1$ indicates a documented interaction between medications i and j. Additionally, we utilize the EHR adjacency matrix $\mathbf{A}_{EHR} \in \mathbb{R}^{|D|\times|M|}$ that encodes the co-occurrence relationships between diagnoses and medications derived from historical electronic health records, providing empirical evidence for appropriate medication-condition associations.

C. Objective Formulation

The goal is to learn a mapping function $f : X^{(1:t)} \rightarrow \mathbb{R}^{|M|}$ that predicts medication probabilities for the current visit based on the patient's historical medical sequence. Formally, given the patient's history up to visit t, we aim to predict:

$$\hat{m}^{(t)} = f(d^{(1:t)}, p^{(1:t)}; \mathbf{A}_{DDI}, \mathbf{A}_{EHR}), \tag{1}$$

where $\hat{m}^{(t)} \in [0,1]^{|M|}$ represents the predicted medication probability vector. The function f must simultaneously optimize multiple competing objectives including therapeutic efficacy, safety constraints, and dosage appropriateness while maintaining computational efficiency for real-time clinical deployment.

The fundamental challenge lies to the trade-offs between recommending medications that are therapeutically effective for the patient's conditions while minimizing the risk of adverse drug interactions and ensuring appropriate dosage levels. This multi-faceted optimization problem requires sophisticated modeling approaches that can capture complex interdependencies between patient history, medication properties, and safety constraints.

IV. Methodology

A. MambaGuard-MOO Framework Overview

Figure 1 illustrates the overall architecture of our MambaGuard-MOO framework, which enhances the MambaHealth foundation with multi-objective optimization and contrastive learning mechanisms. The framework operates through three interconnected modules that work synergistically to address the complex trade-offs in clinical decision-making. The patient health state modeling module processes longitudinal electronic health records through enhanced Mamba blocks that dynamically integrate safety constraints directly into the selective state space computations. The multi-objective prescription generation module employs query attention mechanisms to generate drug recommendations while incorporating comprehensive safety constraints from both DDI and EHR

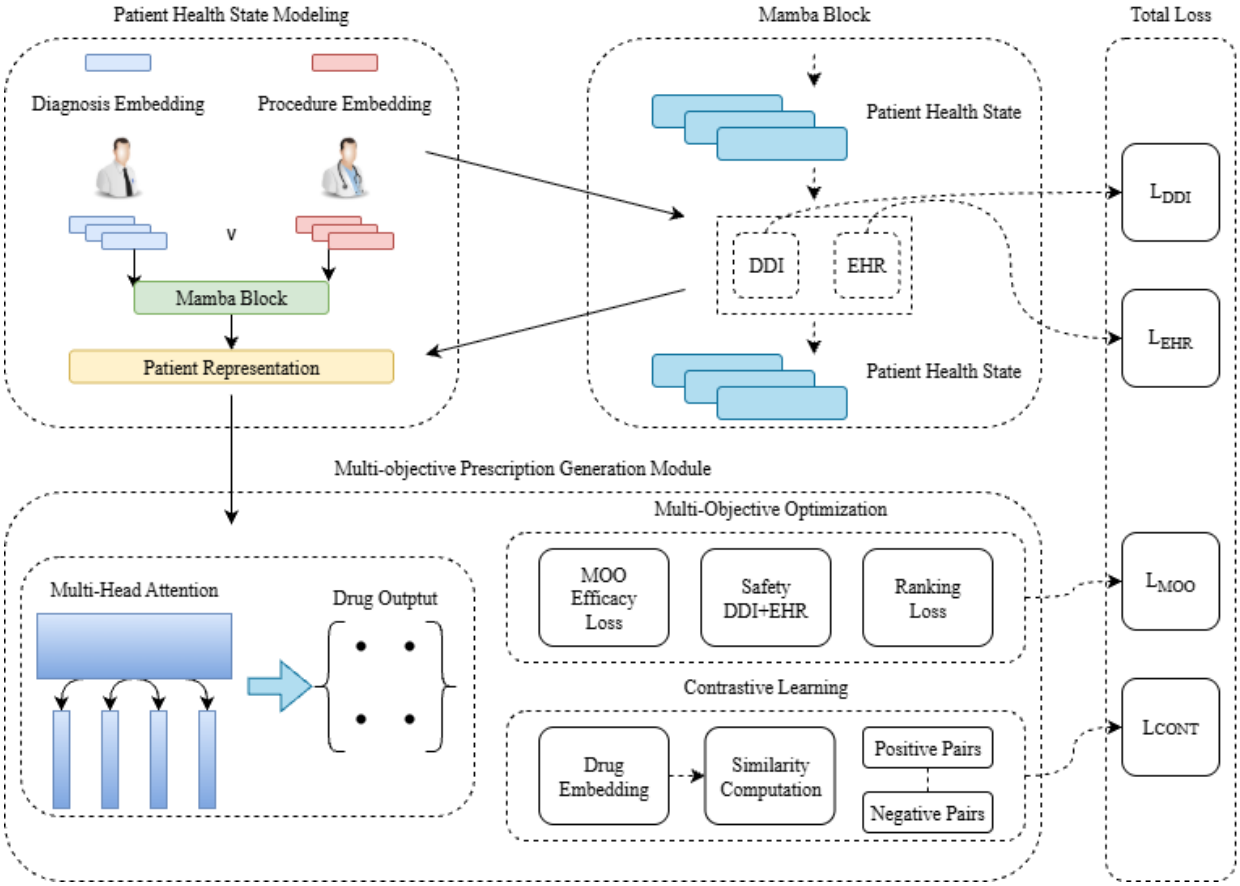


Fig. 1. Overview of the MambaGuard-MOO architecture. The framework consists of three main modules: (1) Patient Health State Modeling with enhanced Mamba blocks that incorporate DDI and EHR constraints into the state transition dynamics, (2) Multi-Objective Prescription Generation Module featuring query attention for drug recommendation and comprehensive safety constraints integration, and (3) Multi-Objective Optimization framework with contrastive learning that balances efficacy, safety, ranking, and personalization objectives through adaptive weighting mechanisms.

sources. Finally, the multi-objective optimization framework with contrastive learning balances competing clinical objectives including therapeutic efficacy, drug interaction safety, ranking precision, and patient personalization through an adaptive weighting scheme that responds to real-time performance feedback.

The key innovation lies in the seamless integration of these components, where the enhanced Mamba architecture provides efficient linear-time processing of patient sequences, the multi-objective framework ensures balanced optimization across competing clinical goals, and the contrastive learning mechanism enhances drug relationship modeling for improving safety and efficacy. This unified approach addresses the fundamental limitations of single-objective methods while maintaining computational efficiency suitable for real-time clinical deployment.

B. Patient Health State Modeling with Mamba Architecture

1) Diagnosis and Procedure Embedding: following the MambaDrug framework, we first transform discrete medical codes into dense representations. For diagnosis information, we design an embedding table $E_d \in \mathbb{R}^{|D| \times \dim}$, where each row corresponds to the embedding vector. Given a multi-hot diagnosis vector $d^{(t)} \in \{0,1\}^{|D|}$, we project the corresponding diagnosis codes into the embedding space:

$$e_d^{(t)} = d^{(t)} \cdot E_d, \tag{2}$$

Similarly, for procedure information, we define a shared embedding table $E_p \in \mathbb{R}^{|P| \times \dim}$ and compute:

$$e_p^{(t)} = d^{(t)} \cdot E_p, \tag{3}$$

Both embedding tables E_d and E_p are learnable parameters shared across all patient visits, enabling the model to capture semantic relationships between medical concepts.

2) Enhanced Selective State Space Modeling: the core innovation of our approach lies in the integration of safety constraints directly into the Mamba architecture's state transition dynamics. We employ State Space Models (SSMs) that dynamically incorporate DDI and EHR information into the model's recurrent computations.

The enhanced SSM formulation operates on the combined patient representation $e_t^{(d,p)} = [e_d^{(t)}, e_p^{(t)}]$ through the following discrete-time state equations.

The state update equation is given by:

$$h_t^{'(d,p)} = A^{(d,p)} h_{t-1}^{(d,p)} + B e_t^{(d,p)}, \tag{4}$$

The output equation is defined as:

$$y_t^{(d,p)} = C h_t^{(d,p)}. \tag{5}$$

where $h_t^{(d,p)}$ is the latent state vector at time t, representing the integrated patient health state. $e_t^{(d,p)}$ is the combined embedding vector for diagnoses and procedures at time t. $y_t^{(d,p)}$ is the output vector at time t, predicting medical actions or interventions.

The adaptive state transition matrix $A^{(d,p)}$ that dynamically incorporates safety and efficacy constraints:

$$A^{(d,p)} = A_0 + W_{\text{DDI}} \cdot \mathbf{D} + W_{\text{EHR}} \cdot \mathbf{E}, \tag{6}$$

where A_0 represents the baseline state transition matrix, $\mathbf{D} \in \{0,1\}^{|M|\times|M|}$ is the DDI adjacency matrix, $\mathbf{E} \in \mathbb{R}^{|D|\times|M|}$ is the EHR-derived medication-diagnosis association matrix, and W_{DDI}, W_{EHR} are learnable weight matrices that modulate the influence of safety and efficacy constraints.

3) Multi-Layer Mamba Processing: our architecture employs multiple Mamba layers to process both diagnostic and procedural information streams. Each Mamba layer incorporates residual connections, layer normalization, and feed-forward networks to enhance the model's representational capacity:

$$\text{MambaLayer}(x) = \text{LN}(\text{Drop}(\text{Mamba}(x)) + x). \tag{7}$$

The separate processing streams for diagnoses and procedures are then concatenated to form a comprehensive patient representation:

$$h_{\text{combined}} = \text{Concat}(h_d^{(T)}, h_p^{(T)}). \tag{8}$$

This combined representation captures the complex interdependencies between a patient's diagnostic history and procedural interventions, providing a rich foundation for subsequent drug recommendation decisions.

C. Multi-Objective Optimization Framework

1) MOO Loss Calculator Architecture: our multi-objective optimization framework addresses the inherent trade-offs in drug recommendation by simultaneously optimizing multiple competing objectives. The MOO Loss Calculator incorporates breakthrough-specific components designed to optimize therapeutic efficacy, safety constraints, dosage appropriateness, and patient-specific factors.

The framework initializes several learnable matrices to model different aspects of drug recommendation:

$$\begin{aligned}\theta_{\text{MOO}} = &\{W_{\text{efficacy}}, W_{\text{safety}}, W_{\text{dosage}}, \\ &W_{\text{compat}}, W_{\text{uncertainty}}\}.\end{aligned} \tag{9}$$

where each component captures specific clinical considerations essential for safe and effective medication recommendations.

2) Efficacy Optimization Loss: the efficacy loss component ensures that recommended medications are therapeutically appropriate for the patient's diagnosed conditions. Given a set of target medications M_{target} and patient diagnoses D_{patient}, the loss of efficacy reads as:

$$\mathcal{L}_{\text{efficacy}} = \text{MSE}(\sigma(W_{\text{efficacy}}[m_i, d_j]), \tau_{\text{efficacy}}), \tag{10}$$

where σ denotes the sigmoid function, $W_{\text{efficacy}} \in \mathbb{R}^{|M|\times|D|}$ models drug-diagnosis efficacy relationships, and τ_{efficacy} represents the target efficacy threshold.

3) Safety Constraint Loss: the safety loss component incorporates both DDI constraints and patient-specific safety considerations. The DDI loss penalizes high-risk drug combinations:

$$\mathcal{L}_{\text{DDI}} = \hat{\mathbf{m}}^T \mathbf{D} \hat{\mathbf{m}}, \tag{11}$$

where $\mathbf{D}$ represents the DDI interaction matrix, and $\hat{\mathbf{m}}$ is the vector of predicted drug probabilities.

Additionally, we incorporate EHR-based safety constraints:

$$\mathcal{L}_{\text{EHR}} = \hat{\mathbf{m}}^{\text{ehr}T} \mathbf{E} \hat{\mathbf{m}}^{\text{ehr}}, \tag{12}$$

where $\mathbf{E}$ captures EHR-derived safety relationships and $\hat{\mathbf{m}}^{\text{ehr}}$ represents EHR-informed medication probabilities.

4) Ranking and Precision-Recall Optimization: to enhance recommendation accuracy, we incorporate ranking-specific losses that optimize the relative ordering of medication candidates:

$$\mathcal{L}_{\text{ranking}} = \sum_{\text{pos}\in S_+} \sum_{\text{neg}\in S_-} \max(0, \gamma - (\hat{m}_{\text{pos}} - \hat{m}_{\text{neg}})), \tag{13}$$

where S_+ and S_- represent positive and negative medication sets respectively, and γ is the ranking margin.

The precision-recall optimization combines precision and recall objectives:

$$\mathcal{L}_{\text{PR}} = \alpha \mathcal{L}_{\text{precision}} + (1-\alpha)\mathcal{L}_{\text{recall}}, \tag{14}$$

where α balances the trade-off between precision and recall optimization.

5) Adaptive Weight Mechanism: the MOO framework employs an adaptive weighting mechanism that dynamically adjusts objective weights based on current performance and clinical context.

$$w_i^{(t)} = w_i^{(0)} \cdot (1 + \beta \cdot \Delta_{\text{performance}}^{(t)}), \tag{15}$$

where $w_i^{(0)}$ represents the initial weight for objective i, β is the adaptation rate, and $\Delta_{\text{performance}}^{(t)}$ captures performance deviation from target metrics.

D. Contrastive Learning Framework

1) Drug Embedding Enhancement: our contrastive learning mechanism enhances drug representations by learning to distinguish between safe and unsafe medication combinations. We maintain drug embeddings $E_{\text{drug}} \in \mathbb{R}^{|M|\times d_{\text{embed}}}$ that are refined through contrastive objectives.

The projection head transforms drug embeddings into a contrastive space:

$$z_i = f_{\text{proj}}(E_{\text{drug}}[i]) = W_{\text{proj}} \cdot \text{ReLU}(W_1 \cdot E_{\text{drug}}[i]), \quad (16)$$

where f_{proj} represents the projection network with learnable parameters W_{proj} and W_1.

2) Contrastive Loss Computation: the contrastive loss encourages similar representations for safe drug combinations while pushing apart unsafe combinations [9].

$$\mathcal{L}_{\text{contrastive}} = \sum_{i,j \in \mathcal{P}} -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k \neq i} \exp(\text{sim}(z_i, z_k)/\tau)}, \quad (17)$$

where $\mathcal{P}$ represents positive pairs (safe combinations), $\text{sim}(\cdot, \cdot)$ computes cosine similarity, and τ is the temperature parameter.

For hard negative sampling, we prioritize drug pairs with moderate DDI risk that are therapeutically similar:

$$\mathcal{L}_{\text{hard_neg}} = \sum_{(i,j) \in \mathcal{H}} \max(0, \mu + \text{sim}(z_i, z_j)), \quad (18)$$

where $\mathcal{H}$ represents hard negative pairs and μ is the margin parameter.

E. Unified Loss Function and Training Procedure

1) Comprehensive Loss Formulation: the complete MambaGuard-MOO loss function integrates all optimization objectives through an adaptive weighting scheme. The unified loss combines base prediction losses with our multi-objective and contrastive learning components:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{base}} + \lambda_{\text{MOO}} \mathcal{L}_{\text{MOO}} + \lambda_{\text{contrastive}} \mathcal{L}_{\text{contrastive}}, \quad (19)$$

where $\mathcal{L}_{\text{base}}$ represents the foundational prediction losses, and λ_{MOO}, $\lambda_{\text{contrastive}}$ are weighting parameters.

The base loss combines binary cross-entropy and multi-label margin losses:

$$\mathcal{L}_{\text{base}} = \alpha \mathcal{L}_{\text{BCE}} + (1 - \alpha) \mathcal{L}_{\text{multi}}, \quad (20)$$

where the binary cross-entropy loss is defined as:

$$\mathcal{L}_{\text{BCE}} = -\sum_{i=1}^{|M|} [m_i \log(\hat{m}_i) + (1 - m_i) \log(1 - \hat{m}_i)], \quad (21)$$

and the multi-label margin loss ensures proper ranking:

$$\mathcal{L}_{\text{multi}} = \sum_{i,j: m_i = 1, m_j = 0} \max(0, 1 - (\hat{m}_i - \hat{m}_j))/|M|. \quad (22)$$

2) Multi-Objective Loss Integration: the MOO loss aggregates all clinical objectives with adaptive weights:

$$\mathcal{L}_{\text{MOO}} = \sum_{k=1}^{K} w_k^{(t)} \mathcal{L}_k, \quad (23)$$

where K represents the number of objectives and $w_k^{(t)}$ are time-dependent weights computed as:

$$w_k^{(t)} = w_k^{(0)} \cdot \max(0, 1 + \rho_k \cdot (\text{DDI}_{\text{current}} - \text{DDI}_{\text{target}})), \quad (24)$$

where ρ_k controls the sensitivity of objective k to DDI rate deviations.

The complete MOO loss encompasses:

$$\begin{aligned} \mathcal{L}_{\text{MOO}} = {} & w_{\text{eff}} \mathcal{L}_{\text{efficacy}} + w_{\text{safe}} \mathcal{L}_{\text{safety}} \\ & + w_{\text{rank}} \mathcal{L}_{\text{ranking}} + w_{\text{PR}} \mathcal{L}_{\text{PR}} \\ & + w_{\text{dose}} \mathcal{L}_{\text{dosage}} + w_{\text{pers}} \mathcal{L}_{\text{personalization}}. \end{aligned} \quad (25)$$

3) Contrastive Learning Integration: the contrastive loss combines positive pair attraction with hard negative repulsion:

$$\begin{aligned} \mathcal{L}_{\text{contrastive}} = {} & \mathcal{L}_{\text{positive}} + \lambda_{\text{hard}} \mathcal{L}_{\text{hard_neg}} \\ & + \lambda_{\text{enhanced}} \mathcal{L}_{\text{enhanced}}, \end{aligned} \quad (26)$$

where $\mathcal{L}_{\text{enhanced}}$ incorporates additional contrastive objectives for improved drug relationship modeling.

4) Dynamic Training Strategy: our training procedure employs a multi-stage approach that progressively emphasizes different objectives:

Stage 1 (Warmup): focus on base prediction accuracy with reduced MOO influence:

$$\lambda_{\text{MOO}}^{(1)} = \lambda_{\text{base}} \cdot \min(1, t/t_{\text{warmup}}), \quad (27)$$

where t represents the current training step and t_{warmup} is the warmup duration.

Stage 2 (MOO Integration): gradually increase MOO component weight while introducing contrastive learning:

$$\lambda_{\text{contrastive}}^{(2)} = \lambda_{\max} \cdot (1 - \exp(-(t - t_{\text{warmup}})/\tau_{\text{ramp}})), \quad (28)$$

where τ_{ramp} controls the ramp-up rate.

Stage 3 (Full Integration): apply complete loss function with performance-based weight adaptation:

$$w_k^{(3)} = w_k^{(2)} \cdot (1 + \xi_k \cdot \text{sigmoid}(\Delta_{\text{metric}}^{(k)})), \quad (29)$$

where ξ_k controls adaptation strength and $\Delta_{\text{metric}}^{(k)}$ measures performance change for objective k.

5) Breakthrough Performance Tracking: the framework maintains performance history to enable breakthrough detection:

$$\begin{aligned} \text{Score}_{\text{breakthrough}} = {} & \omega_1 \cdot J + \omega_2 \cdot \text{PRAUC} \\ & + \omega_3 \cdot (1 - \text{DDI}_{\text{rate}}), \end{aligned} \quad (30)$$

where J represents Jaccard similarity, PRAUC denotes Precision-Recall Area Under Curve, and ω_i are breakthrough-specific weights optimized for ceiling-breaking performance.

The model selection criterion prioritizes breakthrough achievements:

$$\text{Save}_{\text{model}} = \mathbb{I}[\text{Score}_{\text{breakthrough}} > \text{Score}_{\text{best}} \wedge \text{DDI}_{\text{rate}} \leq \tau_{\text{DDI}}], \quad (31)$$

where $\mathbb{I}[\cdot]$ is the indicator function and τ_{DDI} represents the maximum acceptable DDI rate.

V. Experiments and Results

A. Experimental Setup

We conduct comprehensive experiments on the MIMIC-III dataset [13], a widely-used benchmark for drug recommendation systems. The dataset contains longitudinal electronic health records from intensive care unit patients, providing rich temporal information including diagnoses, procedures, and medication prescriptions. Following standard preprocessing procedures from previous work, we filter patients and split the data into training, validation, and test sets.

We evaluate our MambaGuard-MOO framework using four key metrics that capture both effectiveness and safety aspects of drug recommendation. DDI [7] measures the proportion of drug-drug interactions in recommended medication combinations, with lower values indicating safer prescriptions. Jaccard [14] computes the intersection over union between predicted and ground-truth medication sets, evaluating recommendation accuracy. PRAUC [15] assesses the model's ability to rank relevant medications higher than irrelevant ones, particularly important for imbalanced datasets. F1 [16] provides the harmonic mean of precision and recall, offering a balanced measure of recommendation quality. These metrics collectively evaluate the trade-off between therapeutic efficacy and medication safety, which is crucial for clinical deployment.

We compare MambaGuard-MOO against a comprehensive set of baseline approaches representing different methodological paradigms. The baselines include RETAIN [17], LEAP [18], DMNC [19], GAMENet [20], SafeDrug [7], MoleRec [21], SHAPE [22], MambaHealth [23], and CycleTrans [2].

B. Results and Analysis

The experimental results shown in Table I demonstrate the superior performance of our MambaGuard-MOO framework across multiple evaluation metrics. Our approach achieves the best performance in three out of four key metrics, with MambaGuard-MOO obtaining a PRAUC of 0.7676 (2.9% improvement over SafeDrug), Jaccard similarity of 0.5541 (12.6% enhancement), and F1-score of 0.7052 (23.1% improvement over the strongest baseline). While maintaining a competitive DDI rate of 0.0341, our method successfully addresses the fundamental trade-off challenge in drug recommendation by simultaneously optimizing therapeutic efficacy and safety constraints through its innovative multi-objective framework, significantly outperforming traditional approaches like RETAIN and GAMENet as well as modern methods like SafeDrug and MoleRec.

VI. Ablation Study

Table II presents the results, which clearly demonstrate the critical contributions of both components to the overall performance. Ablation studies on MambaGuard-MOO reveal that both the multi-objective optimization (MOO) framework and contrastive learning are critical for optimal performance. Removing MOO significantly worsens safety (DDI rate up 39.3% to 0.0562) and efficacy (PRAUC down 16.2%, Jaccard 22.7%, F1-score 26.0%). Excluding contrastive learning also degrades performance, with DDI rate rising 33.2% to 0.0511 and efficacy metrics dropping 11.5%–28.5%. These results confirm that both components are essential for balancing safety and efficacy in drug recommendation systems.

VII. Conclusion and Future Work

A. Conclusion

This paper presents MambaGuard-MOO, a novel multi-objective optimization framework that enhances drug recommendation systems by addressing the fundamental limitations of existing single-objective approaches. By building upon the MambaDrug architecture and integrating advanced multi-objective optimization with contrastive learning mechanisms, our framework successfully navigates the complex trade-offs inherent in clinical decision-making. The experimental results demonstrate the effectiveness of our approach in achieving superior performance across multiple safety and efficacy objectives, establishing a new paradigm for drug recommendation systems that simultaneously optimizes therapeutic effectiveness while maintaining competitive safety standards.

The success of MambaGuard-MOO lies in its innovative approach to modeling drug recommendation as a true multi-objective problem rather than treating it as a single-objective optimization task. The integration of selective State Space Models with safety constraints directly embedded in the state transition dynamics enables efficient processing of long patient sequences while maintaining linear computational complexity. The contrastive learning mechanism enhances the model's ability to distinguish between safe and unsafe drug combinations through improved drug representations, while the adaptive weighting scheme dynamically balances competing objectives based on clinical context and performance feedback. These technical innovations collectively address the scalability and effectiveness challenges that have limited the practical deployment of previous drug recommendation systems in real-world clinical settings.

B. Future Work

Future research directions encompass several promising avenues for extending and improving our framework. The integration of additional data modalities, such as laboratory results, vital signs, and genetic information, could further enhance personalization and treatment precision through expanded multi-objective formulations. Advanced architectural innovations, including the exploration of hybrid models combining Mamba with transformer components or graph neural networks, may capture even more complex medical relationships and dependencies.

Beyond medication recommendation, our MambaGuard-MOO framework demonstrates significant potential for broader applications in electronic health record (EHR) analysis and diagnosis prediction. Recent advances in EHR-based predictive modeling have shown the transformative potential of deep learning approaches for clinical applications including

TABLE I
PERFORMANCE COMPARISON ON MIMIC-III DATASET

Model	DDI Rate↓	PRAUC↑	Jaccard↑	F1-Score↑
RETAIN(2016)	0.0835±0.003	0.7256±0.004	0.4134±0.001	0.6017±0.005
LEAP(2017)	0.0631±0.003	0.7031±0.006	0.4426±0.008	0.5941±0.001
DMNC(2018)	0.0841±0.003	0.7230±0.003	0.4729±0.004	0.6362±0.002
GAMENet(2019)	0.0874±0.003	0.7162±0.005	0.4672±0.004	0.6383±0.004
SafeDrug(2021)	0.0682±0.004	0.7452±0.006	0.4839±0.002	0.6418±0.004
MoleRec(2023)	0.0741±0.002	0.7275±0.005	0.4482±0.003	0.6054±0.004
SHAPE(2023)	0.0784±0.003	0.7006±0.008	0.4538±0.008	0.6194±0.008
CycleTrans(2024)	**0.0311±0.008**	0.6963±0.025	0.3922±0.085	0.5127±0.001
MambaHealth(2024)	0.0349±0.002	0.7549±0.005	0.4730±0.004	0.6146±0.002
MambaGuard-MOO(Ours)	0.0341±0.001	**0.7676±0.002**	**0.5541±0.002**	**0.7052±0.003**

TABLE II
ABLATION STUDY ON MIMIC-III DATASET

Method	DDI Rate↓	PRAUC↑	Jaccard↑	F1-Score↑
MambaGuard-MOO(Our full version)	**0.0341±0.001**	**0.7676±0.002**	**0.5541±0.002**	**0.7052±0.003**
Without MOO	0.0562±0.004	0.6429±0.005	0.4284±0.003	0.5217±0.005
Without Contrastive Learning	0.0511±0.005	0.6793±0.004	0.4724±0.004	0.5045±0.003

outcome prediction, phenotyping, and treatment response prediction [1], [7]. Our multi-objective optimization approach can be naturally extended to diagnosis prediction tasks, where the framework could simultaneously optimize for diagnostic accuracy, confidence estimation, and clinical interpretability. The selective State Space Model architecture's ability to process long sequential medical histories makes it particularly well-suited for capturing temporal patterns in patient trajectories that are crucial for early disease detection and prognosis.

Furthermore, the contrastive learning mechanisms developed in our work can be adapted for EHR-based phenotyping tasks, where distinguishing between similar patient subgroups is essential for precision medicine. The framework's capacity to handle multi-modal data integration positions it for comprehensive EHR analysis that incorporates structured data (diagnosis codes, laboratory values), unstructured text (clinical notes), and temporal dynamics. This comprehensive approach could significantly advance clinical decision support systems by providing holistic patient risk assessments and treatment recommendations that consider the full spectrum of available clinical information, ultimately improving patient outcomes and healthcare delivery efficiency in real-world clinical settings.

VIII. ACKNOWLEDGEMENTS AND DECLARATION

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